

Statistical Distributions

A Comprehensive Guide

Part A: Theoretical Foundation

Short Notes & Explanations with Visual Diagrams

11 Topics Covered	10 Charts & Diagrams	5+ Key Formulas	Q-Q, PDF, CDF, Distributions & Transforms
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1. What is Statistical Distributions?

Statistical distributions describe how values of a random variable are spread or distributed across a range. They define the **probability** that a variable takes on a particular value or falls within a certain range — making them the backbone of statistical analysis, machine learning, and data science.

Core Components

- **Random Variable (X):** A variable whose value is determined by a random process (e.g., rolling a die, measuring height).
- **Probability:** The likelihood (0 to 1) that X takes a specific value or range.
- **Parameters:** Values that define the shape/location of the distribution (e.g., mean μ , standard deviation σ).

★ Why Do Statistical Distributions Matter?

Statistical distributions are used to:

- Model real-world phenomena (income, height, website traffic)
- Make probabilistic predictions and inferences
- Perform hypothesis testing and confidence intervals
- Drive ML algorithms (Naive Bayes, Gaussian Processes, etc.)

Type	Description
Discrete	Countable outcomes (whole numbers): Binomial, Poisson, Bernoulli
Continuous	Infinite outcomes in a range: Normal, Log-Normal, Exponential
Parameters	Mean (μ), Variance (σ^2), Skewness, Kurtosis shape the distribution
Support	The range of values X can take (e.g., $[0, \infty)$, $(-\infty, +\infty)$, $\{0, 1\}$)

2. What is a Q-Q Plot and Why is it Used?

Q-Q Plot (Quantile-Quantile Plot) is a graphical tool used to determine if a dataset follows a theoretical distribution — most commonly the Normal distribution. It plots the quantiles of sample data against the expected quantiles of a reference distribution.

How to Read a Q-Q Plot

- ✔ **Points fall on the diagonal line:** The data follows the theoretical distribution closely.
- ✗ **Points curve away:** The data deviates (heavy tails, skewness, or different distribution).
- ✗ **S-shaped curve:** Suggests the data has lighter tails than normal (platykurtic).
- ✗ **Reverse S-shape:** Suggests heavier tails than normal (leptokurtic / fat tails).

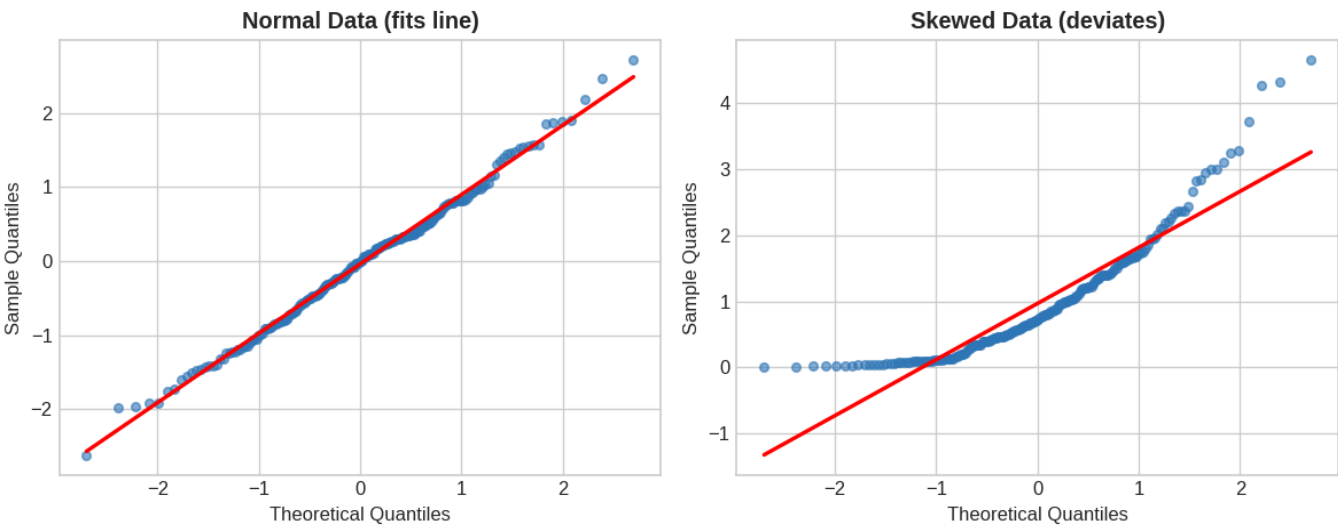


Figure 1: Q-Q Plot — Normal data aligns on the diagonal; skewed data deviates from it

Why is Q-Q Plot Used?

Purpose	Practical Use
Normality Testing	Check if data is normally distributed before applying t-tests, ANOVA
Distribution Fitting	Compare data against any theoretical distribution (exponential, uniform)
Outlier Detection	Extreme deviations at the ends reveal outliers
Transformation Check	Verify if a log or Box-Cox transform normalised the data
Model Diagnostics	Check regression residuals for distributional assumptions

3. Difference Between Discrete and Continuous Distributions

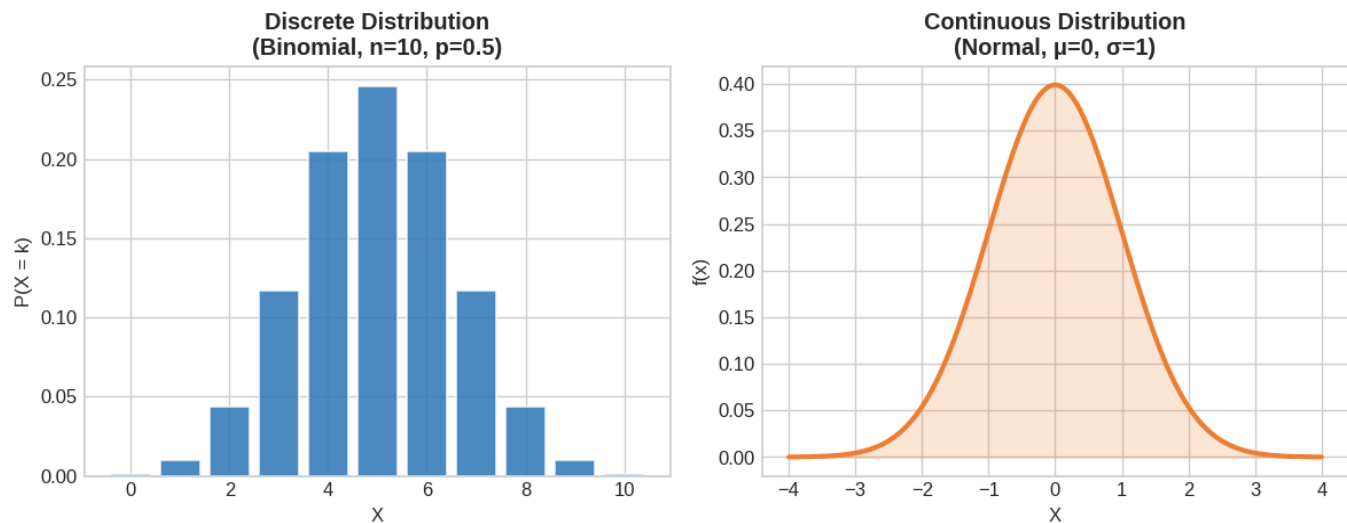


Figure 2: Discrete distribution has isolated bars; continuous has a smooth curve

Feature	Discrete Distribution	Continuous Distribution
Definition	Values are countable / finite	Values can take any real number in a range
Probability	$P(X = k)$ is a specific value	$P(X = x) = 0$; use $P(a \leq X \leq b)$ instead
Function	PMF — Probability Mass Function	PDF — Probability Density Function
Visual	Bar chart (gaps between bars)	Smooth continuous curve
Examples	Binomial, Poisson, Bernoulli	Normal, Log-Normal, Exponential
Real-world	Coin flips, dice rolls, defective items	Height, weight, temperature, time
Sum / Integral	Sum of all PMF values = 1	Integral over full range = 1

4. What is Bernoulli Distribution?

Bernoulli Distribution is the simplest discrete probability distribution. It models a **single trial** with exactly **two outcomes: Success (1) or Failure (0)**. It is the building block of the Binomial distribution.

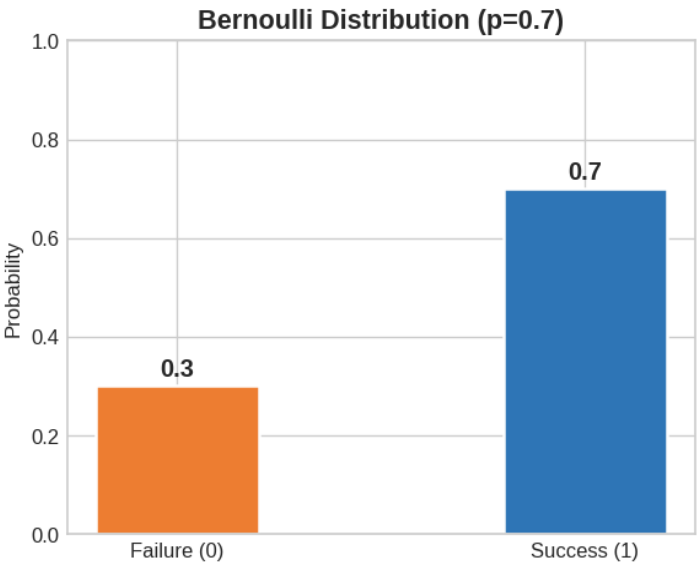


Figure 3: Bernoulli Distribution with $p=0.7$ — $P(X=1) = 0.7$, $P(X=0) = 0.3$

Mathematical Definition

$P(X = k) = p^k \times (1 - p)^{(1-k)}, \quad k \in \{0, 1\}$	PMF
Mean = p Variance = $p(1 - p)$	Statistics

Key Properties

Parameter	p — probability of success ($0 \leq p \leq 1$)
Outcomes	Binary: $X = 1$ (success) or $X = 0$ (failure)
Mean (μ)	p
Variance (σ^2)	$p(1 - p) \rightarrow$ max variance when $p = 0.5$
Special case	$\text{Binomial}(n=1, p) = \text{Bernoulli}(p)$



Real-World Examples

- Email spam detection: Is this email spam? (Yes=1, No=0)
- Medical test: Does the patient have the disease? (Yes=1, No=0)
- Quality control: Is the product defective? (Yes=1, No=0)

5. What is Binomial Distribution?

Binomial Distribution models the number of **successes in n independent Bernoulli trials**, each with the same probability p . It answers: "How many times does an event occur in n repeated experiments?"

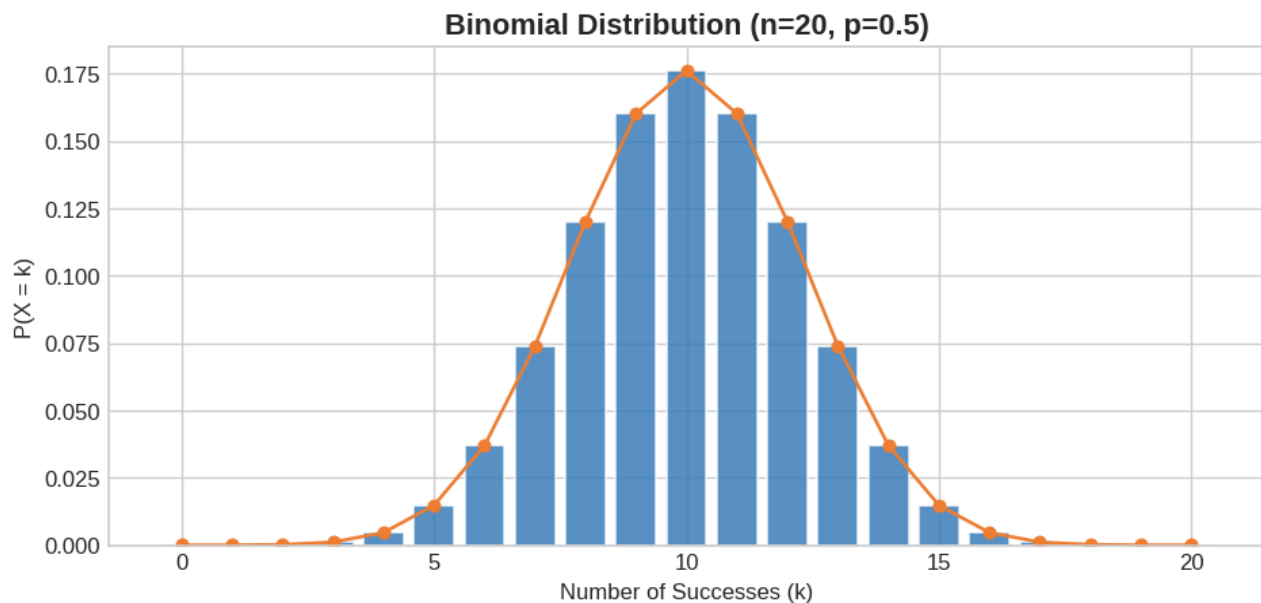


Figure 4: Binomial Distribution ($n=20, p=0.5$) — Bell-shaped, centred at $np=10$

Mathematical Definition

$$P(X = k) = C(n, k) \times p^k \times (1-p)^{n-k}$$

PMF

where $C(n, k) = \frac{n!}{k! \times (n-k)!}$ is the binomial coefficient (number of ways to choose k from n).

$$\text{Mean} = np \quad | \quad \text{Variance} = np(1-p) \quad | \quad \text{Std Dev} = \sqrt{np(1-p)}$$

Conditions for Binomial Distribution

Fixed n	The number of trials is predetermined and fixed
Independence	Each trial is independent of all others
Binary Outcome	Each trial results in exactly Success or Failure
Constant p	Probability of success p remains the same in every trial

Example: Tossing a Coin 10 Times

$n = 10$ trials, $p = 0.5$ (fair coin). What is $P(\text{exactly 6 heads})$?

$$P(X=6) = C(10,6) \times 0.5^6 \times 0.5^4 = 210 \times 0.015625 \times 0.0625 \approx 0.2051 \text{ (about 20.5\%)}$$

6. Log-Normal Distribution

A variable X follows a **Log-Normal Distribution** if its natural logarithm, $\ln(X)$, follows a **Normal Distribution**. It is always positive and right-skewed, making it ideal for modelling quantities that are the product of many small independent factors.

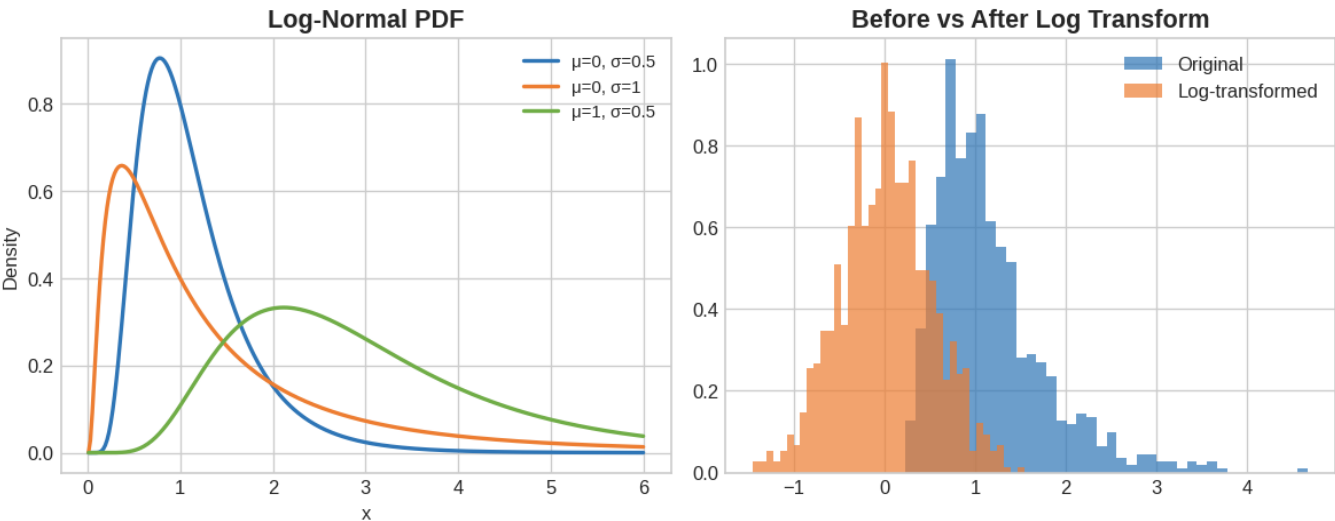


Figure 5: Log-Normal PDF (left) — applying log transform makes data normal (right)

Mathematical Definition

$$f(x) = (1 / (x\sigma\sqrt{2\pi})) \times \exp(-(\ln x - \mu)^2 / (2\sigma^2)) \quad \text{for } x > 0$$

PDF

$$\text{Mean} = \exp(\mu + \sigma^2/2) \quad | \quad \text{Variance} = (\exp(\sigma^2)-1) \times \exp(2\mu+\sigma^2)$$

Key Characteristics

Shape	Always right-skewed (positive skew); never takes negative values
Parameters	μ (mean of $\ln X$) and σ (std dev of $\ln X$)
Log Transform	If $X \sim \text{LogNormal}(\mu, \sigma^2)$, then $\ln(X) \sim \text{Normal}(\mu, \sigma^2)$
Multiplicative	Arises from multiplicative processes (growth rates, ratios)
Heavy Tail	Has heavier right tail than the Normal distribution



Real-World Applications

- Finance: Stock prices, asset returns (multiplicative growth)

- Biology: Body weight, bacterial growth, organ sizes
- Environment: Rainfall, particle size distributions
- Income & Wealth: Income distributions in economics

7. Power Law Distribution

A **Power Law Distribution** (also called Pareto or Zipf distribution) describes phenomena where a relative change in one quantity results in a proportional relative change in another. It captures the **"80/20 Rule"** and **"heavy-tail"** behaviour seen in many natural and social systems.

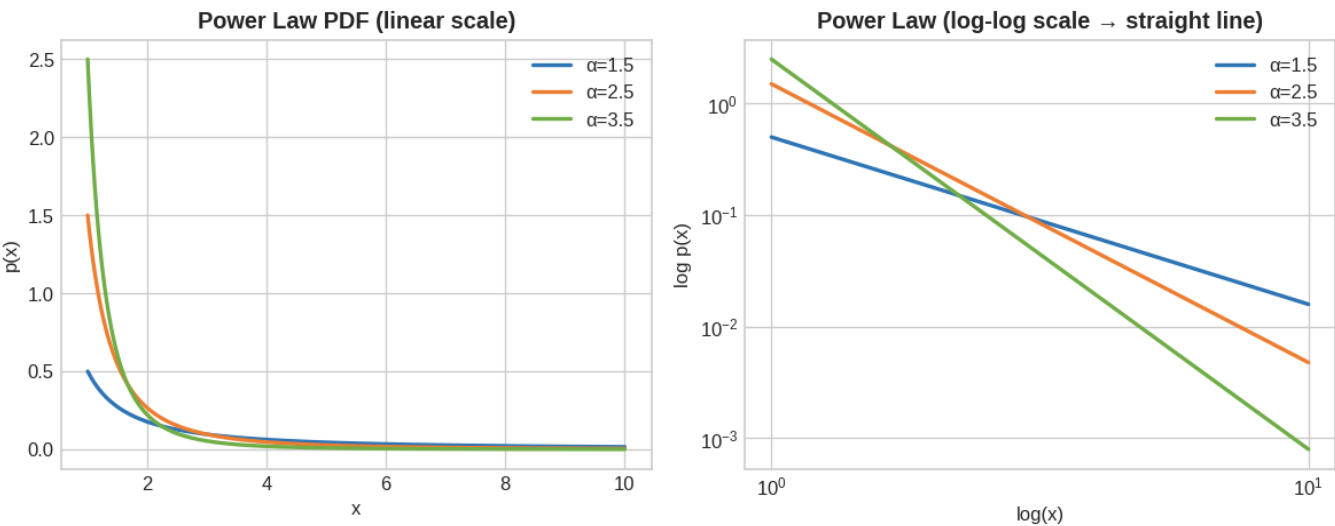


Figure 6: Power Law on linear scale (left) and log-log scale producing a straight line (right)

Mathematical Definition

$$p(x) = (\alpha - 1) \times x^{(-\alpha)} \quad \text{for } x \geq x_{\min}$$

PDF

The key signature: on a **log-log plot**, a power law appears as a **straight line** with slope $-\alpha$.

Key Properties

Exponent α	Controls tail heaviness; typically $2 < \alpha < 3$ in natural systems
Scale-free	No characteristic scale — the distribution looks the same at all scales
Heavy tail	Extreme events are far more common than in a Normal distribution
Log-log linear	Straight line on log-log plot is the diagnostic signature
Mean	Finite only if $\alpha > 2$; Variance finite only if $\alpha > 3$



Real-World Examples of Power Laws

- Wealth Distribution: Top 1% own ~38% of all wealth (Pareto principle)
- Word Frequency: A few words (the, a, is) account for most text usage (Zipf's Law)
- Internet: A few websites (Google, YouTube) get most traffic
- Earthquakes: Gutenberg-Richter law governs earthquake magnitudes

8. What is Box-Cox Transform?

Box-Cox Transformation is a family of power transformations used to **stabilise variance and make data more normally distributed**. It was introduced by George Box and David Cox in 1964. Many statistical methods assume normality — Box-Cox helps meet that assumption for skewed data.

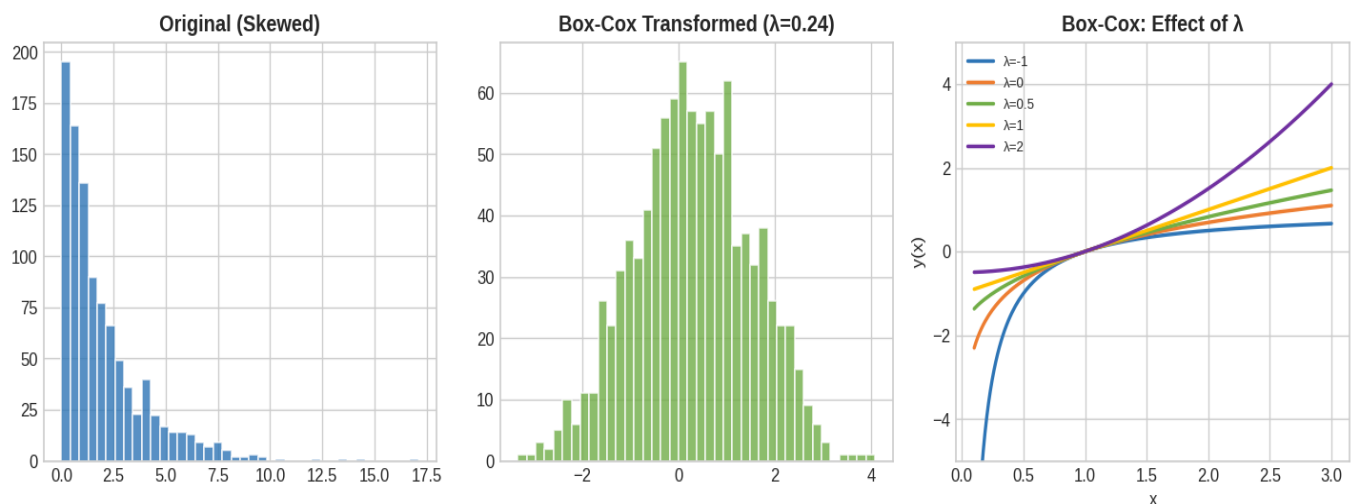


Figure 7: Box-Cox transforms skewed data (left) to near-normal (centre); λ parameter effect (right)

Mathematical Definition

$$y(\lambda) = (x^\lambda - 1) / \lambda \quad \text{if } \lambda \neq 0$$

$$y(\lambda) = \ln(x) \quad \text{if } \lambda = 0$$

λ (lambda) is the transformation parameter, estimated by **Maximum Likelihood Estimation (MLE)** to maximise normality.

Effect of Lambda (λ) Parameter

$\lambda = -1$	Reciprocal transformation: $y = 1/x$
$\lambda = 0$	Natural log transformation: $y = \ln(x)$
$\lambda = 0.5$	Square root transformation: $y = \sqrt{x}$
$\lambda = 1$	No transformation (identity): $y = x$
$\lambda = 2$	Square transformation: $y = x^2$

⚠ Important Notes on Box-Cox Transform

- Only works for strictly positive data ($x > 0$). For zero/negative values, use Yeo-Johnson transform.
- After modelling, predictions must be back-transformed to the original scale.
- In Python: `scipy.stats.boxcox(data)` returns transformed data and optimal λ .

9. Poisson Distribution

Poisson Distribution models the number of events occurring in a **fixed interval of time or space**, given that events occur at a constant average rate (λ) and are **independent** of each other. It is the "counting distribution" for rare events.

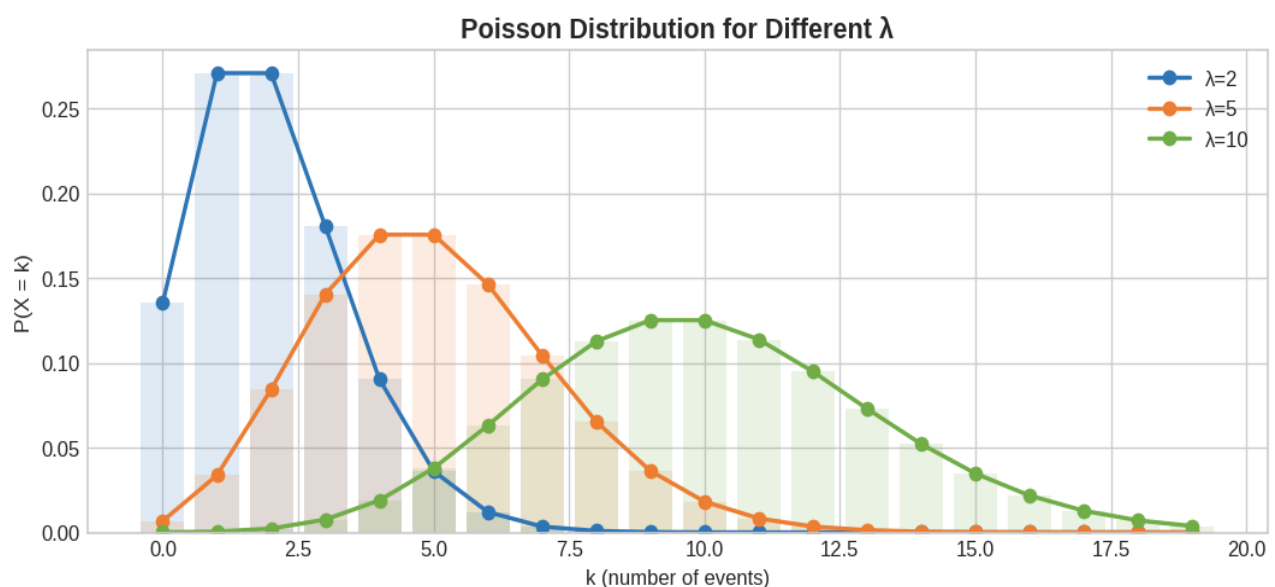


Figure 8: Poisson Distribution for $\lambda=2, 5, 10$ — as λ increases, it approaches Normal shape

Mathematical Definition

$$P(X = k) = (\lambda^k \times e^{-\lambda}) / k! \quad k = 0, 1, 2, 3, \dots$$

PMF

$$\text{Mean} = \lambda \quad | \quad \text{Variance} = \lambda \quad | \quad \text{Std Dev} = \sqrt{\lambda}$$

A unique property: **Mean = Variance = λ** (equidispersion). This is the key diagnostic for Poisson data.

Conditions for Poisson Distribution

Rate constant	Events occur at a constant average rate λ per interval
Independence	Each event is independent of all other events

Single events	Two events cannot occur at exactly the same instant
No upper limit	Number of events k can theoretically be infinite
Rare events	Works best when p is small, n is large (Poisson approximation of Binomial)

Detailed Example: Customer Arrivals

A bank receives an average of $\lambda = 5$ customers per hour. What is the probability of exactly 3 customers arriving in the next hour?

$$P(X = 3) = (5^3 \times e^{-5}) / 3! = (125 \times 0.00674) / 6 = 0.1404$$

So there is about a 14% chance of exactly 3 customers arriving in the next hour.

Real-World Applications

- **Call Centres:** Number of calls per minute.
- **Healthcare:** Number of patients arriving at ER per hour.
- **Manufacturing:** Number of defects per unit of production area.
- **Networks:** Number of HTTP requests per second to a server.
- **Physics:** Radioactive decay events per second (nuclear physics).

10. What is Z-Score Probability?

Z-Score (Standard Score) measures how many **standard deviations** a data point is from the **mean** of its distribution. By standardising, we can use the **Standard Normal Table (Z-table)** to find probabilities for any Normal distribution.

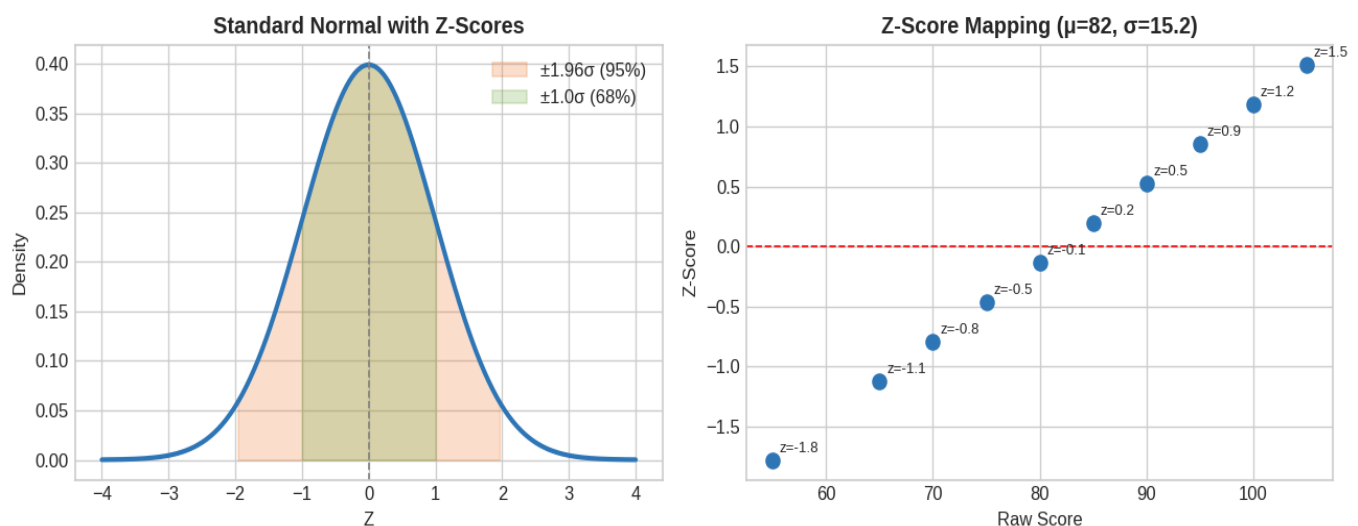


Figure 9: Standard Normal with Z-score regions (left); mapping raw scores to Z-scores (right)

Mathematical Definition

$$Z = (X - \mu) / \sigma$$

Formula

where x = observed value, μ = population mean, σ = standard deviation. Result $Z \sim N(0,1)$.

Z-Score Probability Regions (Empirical Rule)

Z Range	Probability	Coverage	Interpretation
-1 to +1	≈ 68.27%	1σ	Most data points
-1.96 to +1.96	≈ 95%	~2σ	Statistical significance
-2 to +2	≈ 95.45%	2σ	Confidence interval
-3 to +3	≈ 99.73%	3σ	Almost all data
Z > 1.645	≈ 5%	1-tail	Upper 5% critical

 Example: Student Exam Score

A student scores 85 on an exam where $\mu = 70$ and $\sigma = 10$.
 $Z = (85 - 70) / 10 = 1.5$
 $P(Z < 1.5) = 0.9332 \rightarrow$ The student scored better than 93.32% of the class.

11. PDF vs CDF — Probability Density & Cumulative Distribution Functions

PDF and CDF are two fundamental ways to describe a **continuous probability distribution**. They are related: the CDF is the **integral of the PDF**, and the PDF is the **derivative of the CDF**.

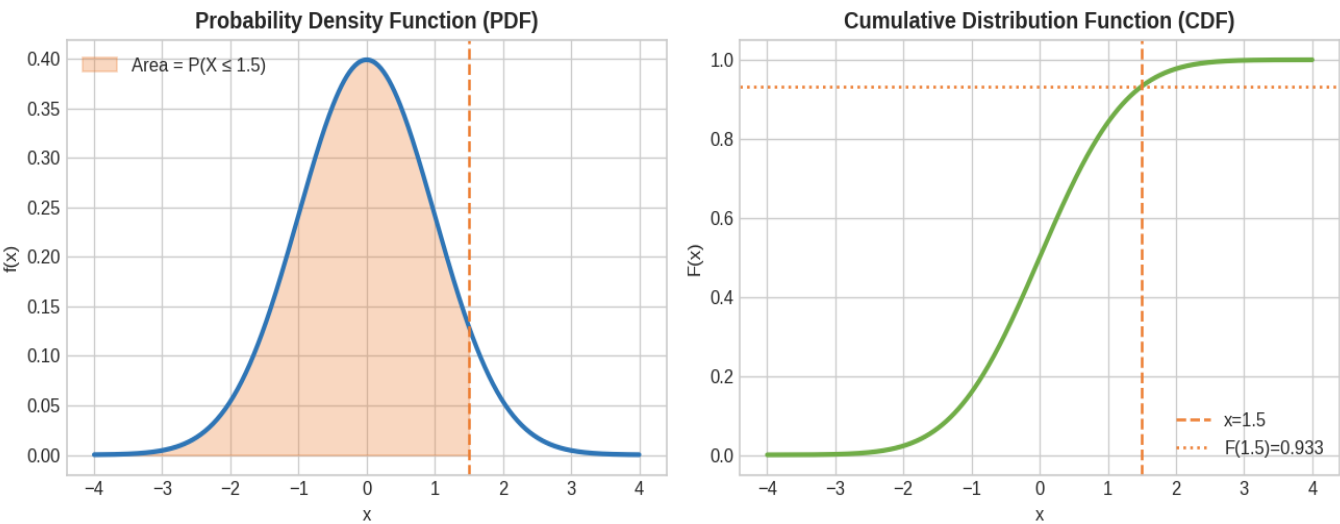


Figure 10: PDF shows density/shape (left); CDF shows cumulative probability (right)

Probability Density Function (PDF)

$f(x) = dF(x)/dx \rightarrow \text{Area under curve between a and b} = P(a \leq X \leq b)$	PDF
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Definition	Describes the relative likelihood of X taking a value near x
Key Property	Area under the entire curve = 1 ($\int f(x)dx = 1$)
Point Probability	$P(X = x) = 0$ for any exact point (area of a line = 0)
Interval Prob.	$P(a \leq X \leq b) = \int [a \text{ to } b] f(x) dx$ (area under curve)
Shape info	Shows where values cluster; peaks = most likely values

Cumulative Distribution Function (CDF)

$F(x) = P(X \leq x) = \int [-\infty \text{ to } x] f(t) dt$	CDF
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Definition	Probability that X is less than or equal to x
Range	Always between 0 and 1; $F(-\infty)=0, F(+\infty)=1$
Monotone	Always non-decreasing (never decreases)
Point Prob.	$P(a \leq X \leq b) = F(b) - F(a)$
Percentiles	The inverse CDF (quantile function) gives percentile values

Aspect	PDF	CDF
Answers	"How dense is probability near x?"	"What fraction is below x?"
Output range	$f(x) \geq 0$ (can be >1)	$0 \leq F(x) \leq 1$
Shape	Bell curve, skewed curve, etc.	S-shaped (sigmoid) curve
Relation	$f(x) = F'(x)$ (derivative of CDF)	$F(x) = \int f(t)dt$ (integral of PDF)
Use case	Understand distribution shape	Compute probabilities and percentiles

Quick Reference Summary

The following table provides a concise overview of all 11 statistical concepts covered in this document:

Concept	Type	Key Parameter(s)	Primary Use / Real-World
Statistical Dist.	Foundation	μ, σ , shape	Modelling random phenomena
Q-Q Plot	Diagnostic	Quantiles	Test normality / distribution fit
Discrete Dist.	Category	Countable X	Coin flips, counts, integer events
Continuous Dist.	Category	Real-valued X	Height, weight, temperature
Bernoulli	Discrete	p	Single binary trial (spam/not spam)
Binomial	Discrete	n, p	n repeated binary trials (quality control)

Log-Normal	Continuous	μ, σ	Income, stock prices, biological sizes
Power Law	Continuous	α (exponent)	Wealth, web traffic, earthquakes
Box-Cox	Transform	λ (lambda)	Normalise skewed data for modelling
Poisson	Discrete	λ (rate)	Event counts per interval (ER arrivals)
Z-Score	Standardisation	μ, σ	Hypothesis testing, probability lookup
PDF	Function	$f(x) \geq 0$	Probability density at a point
CDF	Function	$0 \leq F(x) \leq 1$	Cumulative probability up to x